Architecture of the Microprocessors

Smart House Application Project using Arduino
- Rain Sensor –

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1. Introduction

Overview: One of the most common household inconveniences is having laundry hanging outside to dry, only for unexpected rain to soak the clothes again. This project addresses that issue by creating a "Smart House" application that automatically protects laundry from rain. The objective is to build a system that senses precipitation and physically moves a clothes rack under a roof for protection. Once the rain stops, the system automatically moves the clothes back out to continue drying.

Project Description This project utilizes an Arduino microcontroller to automate the movement of a clothes drying rack on a cardboard house model. The system functions as a continuous feedback loop:

1. **Monitoring:** A rain sensor is positioned to detect falling rain droplets.
2. **Decision Making:** The Arduino continuously reads the sensor data. If moisture is detected (indicating rain), it triggers the protection protocol. If the sensor is dry, it triggers the drying protocol.
3. **Actuation:** A servo motor acts as the mechanical arm.
    - **When Raining:** The servo rotates to drag the clothes rack mechanism *under* the roof of the cardboard house to keep it dry.
    - **When Dry:** The servo rotates in the opposite direction to move the rack *out* from under the roof, exposing the clothes to the air.

Components Used

- **Arduino UNO :** The main controller board.
- **Rain Sensor Module :** Used to detect the presence of water droplets.
- **Servo Motor :** The actuator arm that physically pulls or pushes the clothes rack.
- **Jumper Wires:** For circuit connections.
- **Cardboard House Model:** The physical structure representing the home and shelter.

2. Theoretical Background

2.1. Microcontroller Logic (Arduino) The Arduino UNO is the central processing unit of the system, based on the ATmega328P microcontroller. It serves as the bridge between the physical world (sensors/motors) and the software logic. The Arduino operates by reading electrical states from its Input pins and sending control signals through its Output pins. In this project, it uses digital logic inputs (High/Low) to make decisions and Pulse Width Modulation (PWM) outputs to control movement.

2.2. Resistive Moisture Sensing The rain sensor works on the principle of variable resistance and electrical conductivity.

- **Conductivity:** Pure water is a poor conductor, but rainwater contains electrolytes and impurities that conduct electricity.
- **Operation:** The sensor plate consists of two exposed conductive tracks. When the plate is dry, the electrical path is broken, creating infinite resistance (Open Circuit). When rain falls on the plate, the water bridges the gap between the tracks, drastically lowering the resistance.
- **Digital Output:** An onboard comparator chip (LM393) detects this drop in resistance. When the resistance falls below a set threshold (meaning the plate is wet), the module sends a digital LOW signal (0V) to the Arduino. When dry, it sends a HIGH signal (5V).

3.3. Servo Motor Actuation A servo motor is a precise actuator used for angular positioning. Unlike standard motors that spin continuously, a servo can be told to go to a specific angle (e.g., 0° or 90°) and hold that position against resistance.

- **PWM Control:** The Arduino controls the servo using Pulse Width Modulation. By sending electrical pulses of varying widths (durations), the Arduino tells the internal circuitry of the motor exactly where to position the shaft.
- **Application:** In this project, the servo converts electrical energy into the mechanical force required to drag the friction-based load of the clothes rack.

3. Implementation

3.1. Hardware Connections

The circuit involves three main components: the Arduino, the Rain Sensor, and the Servo Motor. The connections are established as follows:

A. Rain Sensor Module: The sensor sends data to the Arduino regarding the weather status.

- **VCC:** Connected to Arduino **5V** (Power).
- **GND:** Connected to Arduino **GND** (Ground).
- **DO (Digital Output):** Connected to Digital Pin **7**. This pin transmits the logic state (Wet/Dry).

B. Servo Motor: The servo receives commands from the Arduino to move the clothes.

- **Red Wire:** Connected to Arduino **5V**.
- **Brown Wire:** Connected to Arduino **GND**.
- **Orange Wire:** Connected to Digital Pin **6**. This is a PWM-enabled pin required for servo signal control.

3.2. Code Explanation

The software logic is written in C++ within the Arduino IDE.

A. Library and Variable Declaration

```
#include <Servo.h>

Servo s1;

int val = 0 ;
```

- #include <Servo.h>: Imports the dedicated Servo library, which simplifies the complex coding required to generate PWM signals.
- Servo s1: Creates a software object named s1 that corresponds to our physical motor.
- int val = 0: Declares an integer variable to store the data coming from the rain sensor (0 or 1).

B. System Setup

```
void setup()
```

```
{
  Serial.begin(9600);
  pinMode(7,INPUT);  // Rain sensor output pin connected
  s1.attach(6);      // Servo Motor
}
```

- Serial.begin(9600): Starts serial communication, allowing the Arduino to print status updates to the computer screen for debugging.
- pinMode(7, INPUT): Configures Pin 7 to behave as an input, ready to "read" the voltage coming from the rain sensor.
- s1.attach(6): Links the s1 software object to physical Pin 6, enabling control over the motor connected there.

C. Main Operation Loop The loop function runs continuously to monitor weather changes in real-time.

```
void loop()
{
  val = digitalRead(7);  // Rain sensor output pin connected
  Serial.println(val);   // see the value in serial monitor
  delay(100);
```

- val = digitalRead(7): The Arduino checks the state of Pin 7.
    - If val is **0**: It is Raining (Water is conducting electricity on the sensor).
    - If val is **1**: It is Dry.
- Serial.println(val): Displays the current status on the monitor.

D. The "Rain" Scenario (Protection Mode)

```
  if(val == 0 )
  {
    delay(100);
    s1.write(0);   // Servo Rotate to 0 Degrees
  }
```

- If the sensor detects rain (val == 0), the code executes this block.
- s1.write(0): The servo is commanded to rotate to position 0. Mechanically, **this pulls the clothes rack under the roof** to protect it from the rain.

E. The "Dry" Scenario (Drying Mode)

```
  if(val == 1 )
  {
    s1.write(90);  // Servo Rotate to 90 Degrees
  }
}
```

- If the sensor is dry (val == 1), the code executes this block.
- s1.write(90): The servo rotates to position 90 degrees. Mechanically, **this pushes the clothes rack out from under the roof**, exposing the laundry to the sun and wind to dry.

4. Conclusions

The "Smart House Application Project" successfully demonstrates how automation can solve common household problems. By integrating an Arduino, a rain sensor, and a servo motor, the system reliably protects laundry from unexpected rain and automatically exposes it to the sun when the weather clears. This prototype not only offers convenience and peace of mind for homeowners but also promotes energy efficiency by encouraging natural drying over electric dryers. While a full-scale implementation would require stronger motors to handle heavy loads, this project proves that simple, cost-effective engineering can create intelligent systems that significantly improve daily life.

Bibliography

1. Datasheet Arduino
2. Datasheet Rain Sensor
3. Datasheet Servomotor
4. Link YouTube video : Rain Detector project